

Testimony from the Cord Blood Association

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My name is Dr. Joanne Kurtzberg and I am honored to speak on behalf of the Cord Blood Association. I will highlight some of our recommendations for the FDA regarding the proposed guidances and we will also submit more extensive comments to the docket. First, I will review some background information.

I am a pediatric hematologist-oncologist and founded the pediatric blood and marrow transplant program at Duke University which has transplanted >2500 children with malignant and genetic disorders. I founded and direct the Carolinas Cord Blood Bank, an FDA licensed, public cord blood bank. I am the chief scientific and chief medical officer of the Robertson Clinical and Translational Cell Therapy Program at Duke. This program is focused on translating cell therapies to the clinic, and currently conducts phase I and II clinical trials of cord blood cells in children with cerebral palsy, autism and birth asphyxia, and adults with acute ischemic stroke. I direct an FDA registered, GMP cell manufacturing lab, deriving cell therapy products from cord blood and cord tissues. I also co-direct the Stem Cell Transplant Laboratory, a GTP manufacturing, clinical laboratory of Duke Hospital. Thus I have personally experienced the journey from discovery to the clinic. Personally I have had experience manufacturing and transplanting 351 and 361 products, autologous and allogeneic products, minimally manipulated and more than minimally manipulated products, and products used for homologous and non-homologous use.

Today, I speak on behalf of the Cord Blood Association of which I am president. The CBA is a young and vigorous, international, nonprofit organization that promotes the work of the cord blood community for the purpose of saving lives, improving health and changing medicine. CBA members include both public and family banks, industry partners, foundations and individuals in and served by the cord blood community. CBA is currently the only organization solely focused on issues related both public and family cord blood banking.

There is much to learn from the history of cord blood banking and transplantation. Cord blood was first used, in 1988, as a source of HLA-matched, related, donor cells in a 5-year old patient with

Fanconi Anemia undergoing transplantation to treat bone marrow failure. Times were different and the transplant, a first-in-man experiment, performed in a child, with minimal pre-clinical data, was successful. The patient, now 33 years old is living a normal life, 27 years later. Importantly, his blood and immune systems are fully comprised of his sister's cord blood cells.

This transplant paved the way for the whole field of cord blood transplantation. Today, there have been more than 35,000 cord blood transplants performed around the world. Over 160 cord blood banks have been established. The public, unrelated donor inventory is estimated to be >700,000 units and >4 million privately banked units worldwide.

Cord blood was the first hematopoietic stem cell product to be formally regulated by the FDA. This was, at least in part, stimulated by a legislative mandate in the Stem Cell Act of 2005 which established the CW Bill Young Cell Transplantation Program administered under HRSA. To date, 7 public cord blood banks have successfully completed the BLA process.

The BLA process for cord blood has been very challenging for the public banking industry, increasing costs without clear evidence of improving the quality of the product. Gaps in the current FDA pathway for approval of cell therapy products were realized. A few of many examples are:

- 1. Cells, unlike drugs, do not expire. The shelf-life or expiration date is not known.**
- 2. Stability protocols, performed to extend expiration dates, sacrifice unique cell products that cannot be replaced.**
- 3. Excessive environmental monitoring adds little, if any, value to manufacturing that is performed in a closed system when appropriate qualification testing is performed and specifications are met.**

4. The delivery of babies, although sanctioned by nature, is not sterile, not controlled, and is a highly variable process. Cord blood and cord tissue are sourced from this disadvantaged position. Evidence-based flexibility is critical in this situation to enable the use of these valuable products.

Today, in the United States, public cord blood banks operate under BLA and or IND. They are accredited by FACT or AABB.

Products are distributed through the National Marrow Donor Program. Outcomes of transplants are reported to the Stem Cell Transplant Outcomes Database, a contract held by the Center for International Blood and Marrow Transplant Research. *It is important to note that the current BLA pathway is solely for unrelated use, and regulatory pathways for private/family banks to provide cord blood and cord tissue for autologous and or family use, do not currently exist.*

Cord blood and cord tissue-derived products have enormous potential for the development of novel cell-based therapies that will have a critical role in the fields of cellular therapies and regenerative medicine. In the view of the CBA, the proposed guidances should be modified to further enable these therapies to reach the clinic.

The CBA would first like to emphasize the following high-level points:

1. Cord blood is not a bag of stem cells. While it does contain small numbers of blood stem cells capable of restoring hematopoiesis after myeloablative insults, the majority of cells are differentiated blood cells. Some of the other cells have therapeutic value but do not act through engraftment, tissue integration or differentiation. Rather they are effector cells acting through paracrine or trophic signaling. As such, we strongly encourage the FDA to consider these mechanisms of action as homologous.
2. The current regulatory framework, which is largely focused on review of drugs, is not sufficient for review of

cellular therapies. We encourage the FDA to modify these regulations to address the unique properties of cells.

3. The designation of minimal or more than minimal manipulation should be risk-based, with consideration of disease-based need, route of administration and the complexity of manufacturing of the specific product. The CBA suggests that if the cells are prepared aseptically and only exposed to FDA approved-for-human-use reagents and or devices, and administered via a less risky route, manufacturing should be considered minimally manipulated. On the other hand, if reagents or devices that are not approved for human use are used in manufacturing, or the route of administration is invasive, the product should be considered more than minimally manipulated.
4. The distinction between function and donor source throughout these guidances is confusing. The designation of 1271 products including autologous cells or tissue as well as cells and tissues from first and second degree relatives is outdated. If HLA match is the operative in this reasoning, then the guidance should state that related HLA identical or haplo-identical products are covered.
5. The FDA should consider a pathway for cellular therapies similar to that already established for hematopoietic stem cell and solid organ transplantation. Therapies could be monitored under IND and outcomes could be reported to a registry such as the CIBMTR already established through HRSA. Expanded access studies could also be used to monitor safety.

The CBA has the following specific comments related to 2 of the guidances under discussion today:

1. *Human Cells, Tissues, and Cellular and Tissue-Based Products (HCT/Ps) from Adipose Tissue: Regulatory Considerations*

This guidance doesn't acknowledge mesenchymal stromal cells (MSCs), the primary cell therapy extract from Adipose tissue. These cells represent a major therapeutic resource and should be considered homologous when used to exert paracrine effects, e.g. to modulate inflammation and autoimmunity or to induce tolerance to donor derived solid organs transplanted to restore function. This has relevance not only to MSC derived from adipose tissue, but MSC from cord tissue and bone marrow.

I will end with comments about the Homologous use guidance.

2. Homologous Use of Human Cells, Tissues, and Cellular Tissue-Based Products:

This guidance is particularly relevant for cord blood bankers and for patients who may benefit from autologous and allogeneic cord blood therapies extending beyond use for hematopoietic reconstitution. An example would be the treatment of young children with cerebral palsy with autologous cord blood. This is a low-risk, outpatient procedure, with encouraging safety and efficacy results in Phase I and Phase II trials. The mechanism of action, as demonstrated in preclinical murine and rabbit experiments, is trophic through paracrine signaling.

In the draft guidance for “Homologous Use of Human Cells, Tissues, and Cellular and Tissue-Based Products,” FDA states in section 3-1.c: “A manufacturer provides HPCs derived from cord blood with a package insert stating that cord blood may be infused intravenously to differentiate in neuronal cells for treatment of cerebral palsy. This is not homologous use because there is insufficient evidence to support that such differentiation is a basic function of these cells in the donor.”

In this instance, the CBA feels that FDA incorrectly assumes that the mechanism of action is through integration of cord blood “stem” cells capable of differentiating into neuronal cells, which we agree would be non-homologous use. However, autologous cord blood cells are acting through signaling mechanisms that are innate properties of the infused cells and that act on endogenous cells in the patient through homologous mechanisms. An example of this

would be signaling by CD14 cells, which make up approximately 10% of the cells in a typical cord blood unit. These cells, which may be of placental origin, have gene expression profiles and biologic activities that are very different from adult, bone marrow derived, CD14 cells. These cells have been shown to protect neural cells from hypoxic insults in organotypic and animal models as well as to be source of *ex vivo* expanded microglial-like cells capable of inducing oligodendrocyte proliferation and promoting re-myelination in animal models. These cells are currently undergoing testing in a phase I clinical trial in pediatric patients with leukodystrophies. Over 200 clinical trials utilizing cord blood or cord tissue derived cells are currently listed on <clinicaltrials.gov>.

So we have an autologous, not more than minimally manipulated, product used for homologous? Or non-homologous use. Let's consider the consequences of this decision.

If the FDA accepts the rationale that biological activities of cord blood cells are homologous, then administration of autologous cord blood which is not more than minimally manipulated, would be viewed as practice of medicine and regulated under 1271/361. However, if the FDA does not agree with this argument and expects these products to be licensed under a BLA, who gets the BLA? Does each family/private bank, currently non-regulated entities, go through the BLA process? Does the treating institution obtain the BLA? Does a public bank get the BLA? How are international products handled? The list of questions goes on and on....and the CBA welcomes the opportunity to engage in meaningful conversation with the FDA regarding these questions.

The CBA is committed to bringing effective cord blood and cord tissue derived therapies to patients as safely and efficiently as possible. Clarification of some of the issues raised in our comments will help move the process forward.

The CBA thanks the FDA for the opportunity to raise these issues. We look forward to the FDA's feedback on our comments

References:

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